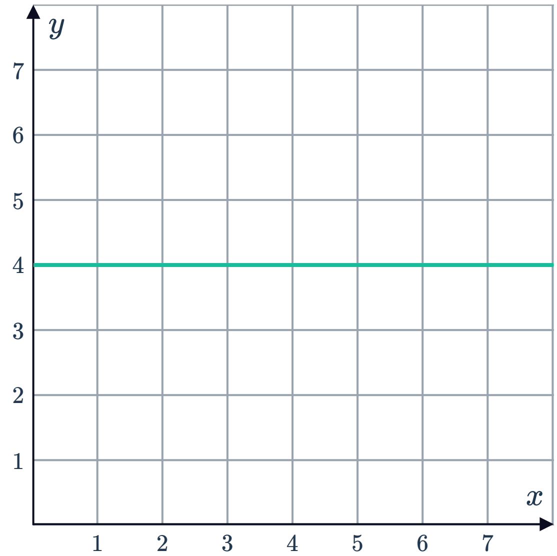
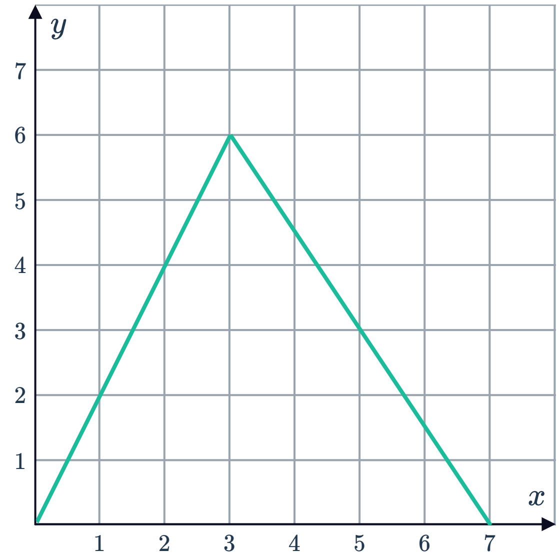
Geometric areas

1 Calculate geometrically, the area bounded by the following functions and the x-axis over the given domain:

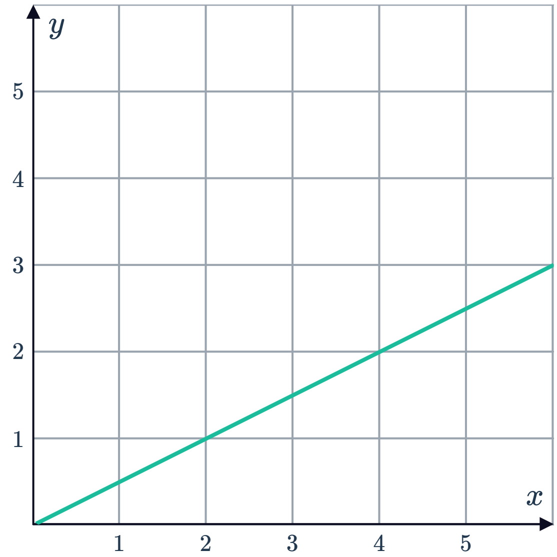
a $2 \leq x \leq 5$



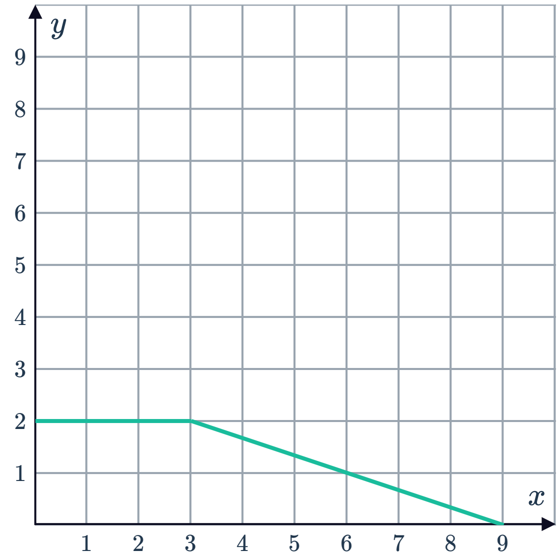
b $0 \leq x \leq 7$



c $0 \leq x \leq 4$

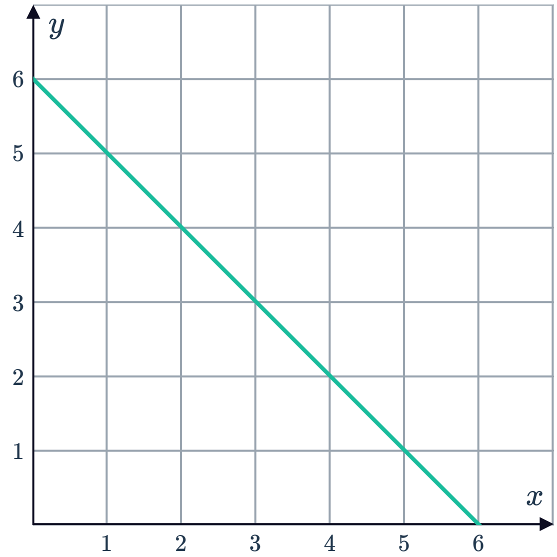


d $0 \leq x \leq 9$

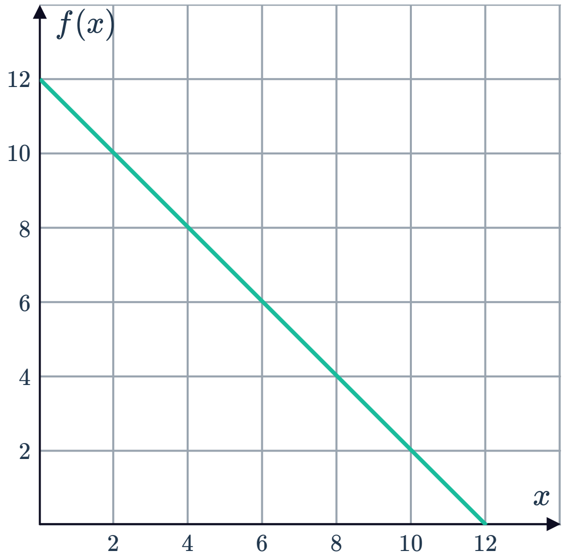


2 Calculate geometrically, the exact value o[redacted]ollowing definite integrals:

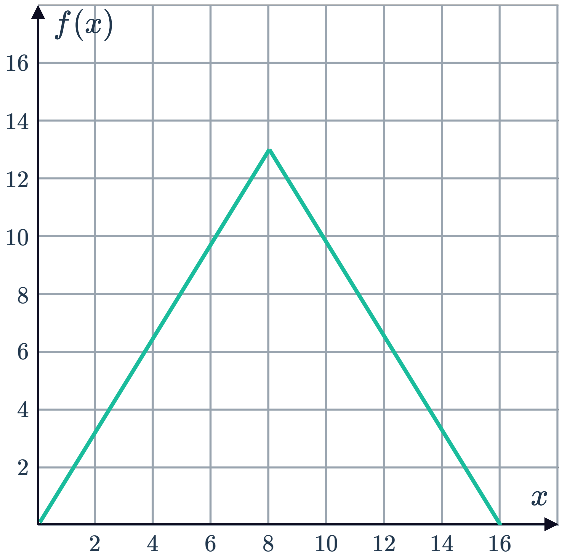
a $\int_0^6 (6-x)\,dx$



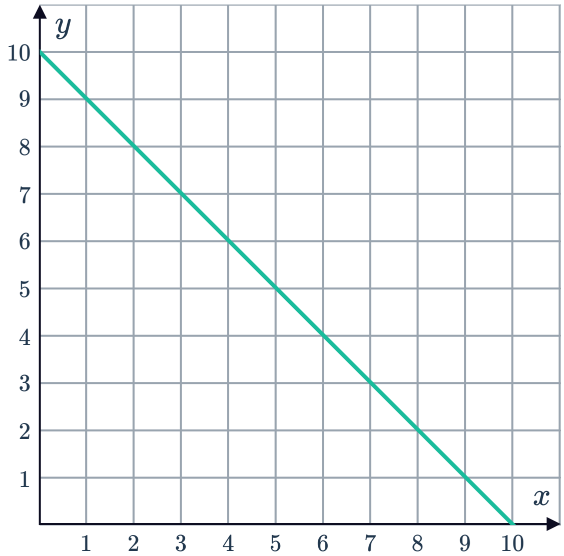
b $\int_0^{12} f(x)\,dx$



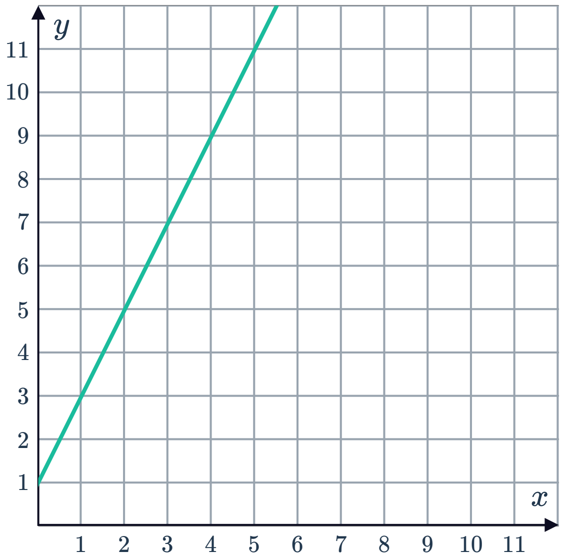
c $\int_0^{16} f(x)\,dx$



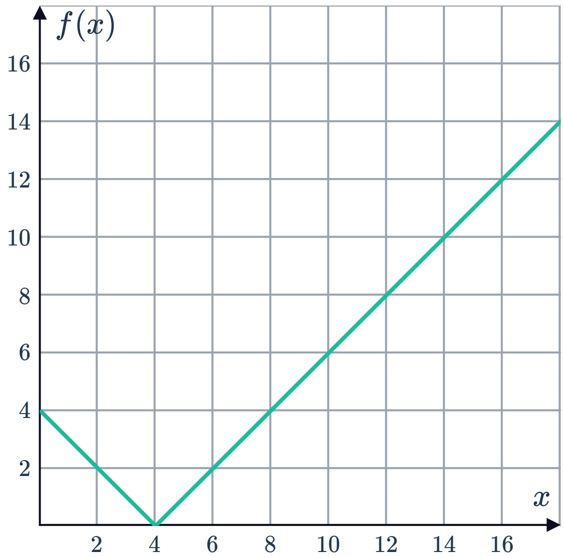
d $\int_1^8 (10-x)\,dx$



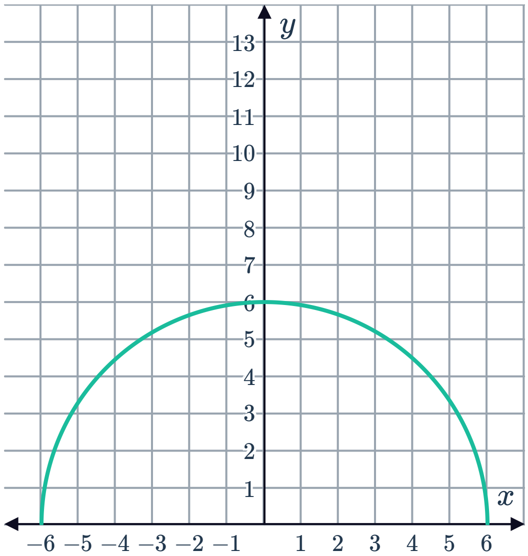
e $\int_3^5 (2x+1)\,dx$



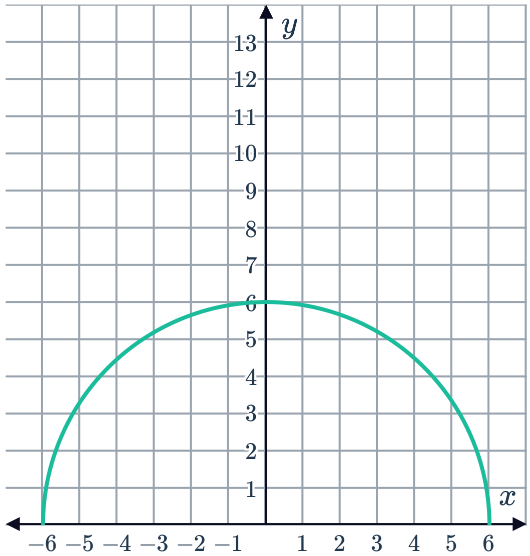
f $\int_0^{16} f(x)\,dx$



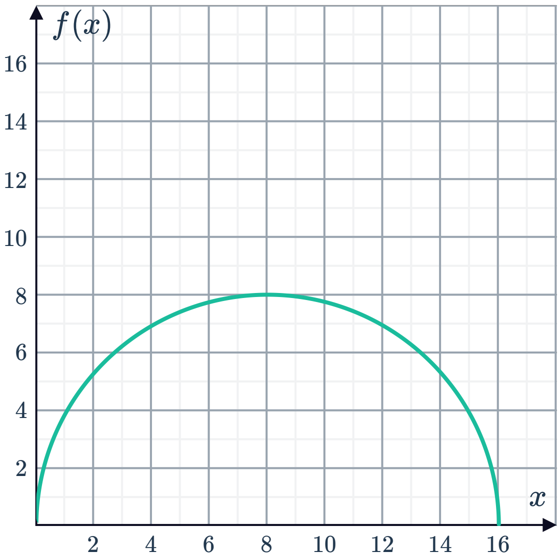
g $\int_{-6}^{6} \sqrt{36 - x^2}\, dx$



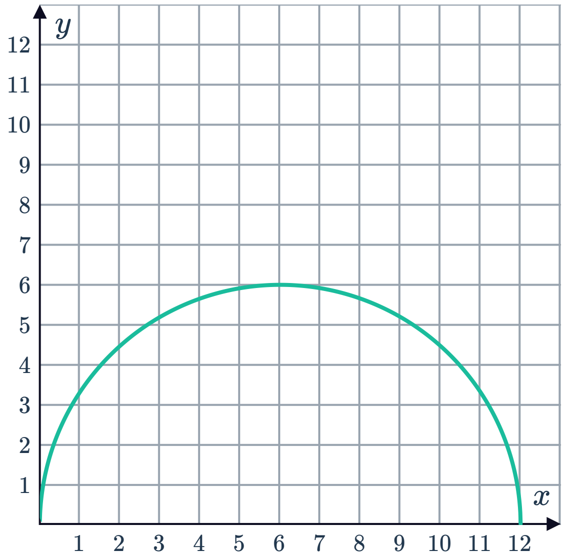
h $\int_{0}^{6} \sqrt{36 - x^2}\, dx$



i $\int_{0}^{16} f(x)\, dx$



j $\int_{0}^{12} \sqrt{36 - (x-6)^2}\, dx$



3 For each of the following functions:

i Sketch a graph of $f(x)$.

ii Hence, calculate geometrically the area bounded by the curve and the x-axis over the given domain:

a $f(x) = 6 - 6x$ for $0 \leq x \leq 1$

b $f(x) = 3x + 4$ for $0 \leq x \leq 3$

c $f(x) = \begin{cases} x & \text{for } 0 \leq x < 4 \\ 8 - x & \text{for } 4 \leq x \leq 8 \end{cases}$

d $f(x) = \begin{cases} 2x & \text{for } 0 \leq x \leq 3 \\ 6 & \text{for } 3 < x < 6 \\ 18 - 2x & \text{for } 6 \leq x \leq 9 \end{cases}$

Approximate areas

4 Consider the function $f\left(x\right)=x^2-3x+10$.
For each of the following intervals, state whether the right endpoint approximation method will underestimate or overestimate the area under the curve. Explain your answer.

a $[3,8]$

b $[-6,-2]$

5 Consider the function $f\left(x\right)=x^2+5$.
For each of the following intervals, state whether the left endpoint approximation method will underestimate or overestimate the area under the curve. Explain your answer.

a $[2,6]$

b $[-5,-1]$

6 Consider the function $f\left(x\right)=10e^x$.
State the approximation method that will give an underestimate of the true area on any given interval. Explain your answer.

7 Consider the function $f\left(x\right)=-3x^2-24x+1$.

a State whether the right endpoint approximation to the area will overestimate or underestimate on the following intervals.

i $[-10,-5]$ ii $[-4,2]$ iii $[0,2]$ iv $[-6,-4]$

b State whether the left endpoint approximation to the area will overestimate or underestimate on the following intervals.

i $[-10,-5]$ ii $[-4,2]$ iii $[0,2]$ iv $[-6,-4]$

8 The function $f\left(x\right)=5x$ is defined on the interval $[0,6]$.

a Graph $f\left(x\right)$.

b Find the area under $f\left(x\right)$ by partitioning $[0,6]$ into 3 sub-intervals of equal length using:

i Left endpoint approximation

ii Right endpoint approximation

c Find the area under $f\left(x\right)$ by partitioning $[0,6]$ into 6 sub-intervals of equal length using:

i Left endpoint approximation

ii Right endpoint approximation

d Find the actual area under the curve on the interval $[0,6]$.

9 The function $f(x) = -4x + 12$ is defined [redacted] interval $[0, 3]$.

a Graph $f(x)$.

b Find the area under $f(x)$ by partitioning $[0, 3]$ into 3 sub-intervals of equal length using:

i Left endpoint approximation

ii Right endpoint approximation

c Find the area under $f(x)$ by partitioning $[0, 3]$ into 6 sub-intervals of equal length using:

i Left endpoint approximation

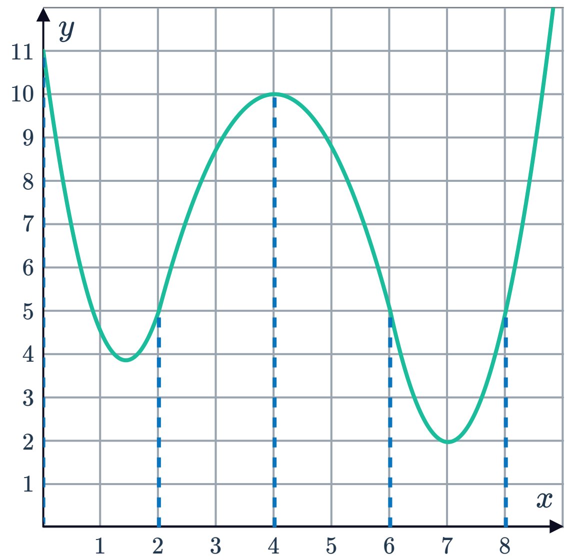
ii Right endpoint approximation

d Find the actual area under the curve on the interval $[0, 3]$.

10 The interval $[0, 8]$ is partitioned into 4 sub-intervals $[0, 2]$, $[2, 4]$, $[4, 6]$, and $[6, 8]$. Find the area under the curve on the interval $[0, 8]$ using:

a Left endpoint approximation

b Right endpoint approximation



11 Use left endpoint approximation to find $\int_1^9 2x^2 dx$ by using 4 rectangles of equal width.

12 Use right endpoint approximation to find $\int_1^5 \frac{1}{x} dx$ by using 4 rectangles of equal width. Give your answer as a simplified fraction.

13 Approximate $\int_3^{15} (4x + 6)\, dx$ by using 4 rectangles of equal width and using the method:

a Left endpoint approximation

b Right endpoint approximation

Use technology

14 Use technology to find the exact value of the following definite integrals:

a $\int_2^5 2x + 3\ dx$

b $\int_0^3 e^x\ dx$

c $\int_{-3}^3 -x^2 + 9\ dx$

15 Consider the function $f(x) = 0.5x^2$.

a Complete the table to estimate the area between the function and the x-axis for $1 \le x \le 3$ using the left endpoint approximation method. Round your answers to three decimal places.

n	5	10	100	1000	10 000
A_L units^2					

b Use technology to evaluate $\int_1^3 0.5x^2\, dx$ and hence confirm that the exact area is the limit of A_L as n gets larger.

16 Consider the function $f(x) = \sqrt{4 - x^2}$.

a Complete the table to estimate the area between the function and the x-axis for $0 \le x \le 2$ using the right endpoint approximation method. Round your answers to five decimal places.

n	5	10	100	1000
A_R units^2				

b Use technology to evaluate $\int_0^2 \sqrt{4 - x^2}\, dx$ and hence confirm that the exact area is the limit of A_R as n gets larger.

17 Consider the function $f(x) = -x^2 + 3x$.

a Sketch a graph of $f(x)$.

b State the x-values that define the region bounded by the curve and the x-axis. Write your answer as an inequality.

c Complete the table to estimate the area of the region bounded by the function and the x-axis using the right endpoint approximation method. Round your answers to four decimal places.

n	5	10	100	1000	10 000
A_R units^2					

d Use technology to confirm that the exact area is the limit of A_R as n gets larger.